

Stable Operation of AlGaIn/GaN HEMTs for 25 hours at 400°C in air

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Abstract—Extreme environments such as the Venus atmosphere are among the emerging applications that demand electronics that can withstand high-temperature oxidizing conditions. While wide-bandgap technologies for integrated electronics have been developed so far, they either suffer from gate oxide and threshold voltage (V_{th}) degradation over temperature, large power supply requirements, or intrinsic base current. In this letter, AlGaIn/GaN high electron mobility transistors (HEMTs) are suggested as an alternative platform for integrated sensors and analog circuits in extreme environments in oxidizing air atmosphere over a wide temperature range from 22°C to 400°C. An optimal biasing region, with a peak of transconductance ($g_{m,peak}$) at -2.3 V with a negligible shift over the temperature range was observed. Moreover, remarkably low V_{th} variation of 0.9% was observed, enabling the design of analog circuits that can operate over the entire temperature range. Finally, the operation of the devices at 400°C and 500°C over 25 hours was experimentally studied, demonstrating the stability of the DC characteristics after the 5 hours of burn-in, at 400°C.

Index Terms—Gallium nitride (GaN), High electron mobility transistor (HEMT), High-temperature electronics.

I. INTRODUCTION

WIDE-bandgap semiconductors such as silicon carbide (SiC) and gallium nitride (GaN) have proven to be viable candidates for operation in extreme environments due to their superior electronic properties, namely low intrinsic carrier concentration [1], [2]. Wide-bandgap electronics have recently found use in aviation, space exploration, automotive and deep-well drilling applications. NASA Glenn Research center (GRC) has studied SiC JFETs (junction field effect transistors) for harsh environments for approximately a decade [3] and demonstrated reliable electronics operational at 500°C for one year [4]. High-temperature devices and circuits in SiC have also been studied in n-type metal-oxide semiconductor (NMOS) and complementary metal-oxide semiconductor (CMOS) [5]–[8], and BJT (bipolar junction transistor) [9]–[18]. The high-temperature SiC MOSFETs suffer from gate oxide and V_{th} degradation, while SiC JFETs demand large power supplies and SiC BJTs need a constant base current to operate. To address the need for low-power and high-temperature electronics, AlGaIn/GaN high electron mobility transistor (HEMTs) are suggested.

AlGaIn/GaN HEMTs are a promising candidate for implementation of integrated electronics [19]–[22], and a few studies have discussed the high-temperature capability of the HEMTs [23]–[25]. However, as compared with SiC, AlGaIn/GaN HEMTs have not been sufficiently matured for implementation of integrated circuits (ICs) for extreme temperatures. Previous work has investigated the high-temperature properties of AlGaIn/GaN HEMTs, at temperatures of 200°C [26], 400°C [27], and 425°C [28]. Maier et al. [24] reported the failure of the GaN HEMTs with $Al_{0.24}Ga_{0.76}N$ barrier layer grown on SiC, with Ti/Al/Ni/Au Ohmic contacts and Mo/Au Schottky gate contact after 219 hours at 500°C in vacuum (benign environment). The corresponding failure was associated with the gate diode breakdown. In [29], among other tests, temperature step stress test has been conducted on AlGaIn/GaN to study the reliability of the devices until failure. However due to the nature of the reliability test, limited discussions on the variation of the DC characteristics over temperature has been provided. Furthermore, the DC characteristics of AlGaIn/GaN HEMTs and MIS-HEMTs have been studied up to 600°C in air for 30 minutes and the HEMTs fail prematurely at 300°C due to the high gate leakage [30]. The MIS-HEMTs operated up to 600°C, however, exhibit (V_{th}) instability beyond 300°C. Besides, formation of bubbles is observed above 300°C on the Ni/Au gates. In another study, the gate leakage in Pd/Au gates has shown to be lower than WSi, Ni, and Ir-based gates [31]. This is especially crucial at elevated temperatures, where the gate leakage tends to exacerbate.

In this letter, we investigate high-temperature operation of AlGaIn/GaN HEMTs on Si substrates for analog ICs from 22°C up to 400°C. It is shown that the AlGaIn/GaN HEMTs exposed to 400°C environments have a stable response over 25 hours of operation with $\sim 1\%$ variation of V_{th} after 5-hours burn-in. Moreover, we show that the prolonged exposure to temperatures above 400°C in air can limit the operation of the AlGaIn/GaN HEMTs due to the reduction of the 2DEG (2D electron gas) sheet density and mobility.

II. ALGAIN/GAN HEMT FABRICATION

The HEMTs were fabricated with an AlGaIn/GaN-on-Si wafer (DOWA, Inc.) grown by metal-organic chemical vapor deposition (MOCVD). The cross-section is shown in Fig. 1. It consists of a 1.5 μm buffer structure followed by a 1.5 μm thick GaN layer grown on top of Si (111). Formation of the 2DEG was accomplished by growing an epitaxial

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